

Chapter 31

Domain Range Undefined

1	2	3	4	5	6	7	8	9	10
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1) All of the following values of x make the function $f(x) = \sqrt{x-7}$ undefined **except**?

- A) -4 B) -1 C) 1 D) 5 E) 7

2) Which value of x makes the function $g(x) = \frac{1}{x-5}$ undefined?

- A) -4 B) -1 C) 1 D) 5 E) 7

3) If the Range of a set of points consists of all positive numbers and the Domain of that set of points consists of all negative numbers, **which quadrant or quadrants will the set of all points be located in?**

- A) I only B) I & II C) II only D) III only E) I, III & IV

4) What is the Domain of the following function?

$$h(x) = \sqrt{\frac{x^2 - 5x + 6}{x - 2}}$$

- A) $\{x \mid x \in \mathbb{R}\}$
B) $\{x \mid x \neq 2\}$
C) $\{x \mid x \geq 3\}$
D) $\{x \mid x < 2\}$
E) $\{x \mid x \leq -2\}$

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5) If the **Domain** of $m(x)$ is $\{x \mid x \neq 7\}$, which of the following could be the equation of the function?

A) $m(x) = 3x - 2$

B) $m(x) = \frac{12}{x+7}$

C) $m(x) = \frac{12}{x-7}$

D) $m(x) = \frac{1}{x+7}$

E) $m(x) = 3x + 2$

6) What is the domain of the following function?

$$h(x) = \sqrt{\frac{1}{(x+7)(x+3)(x-2)}}$$

A) $\{x \mid x \in \mathbb{R}\}$

B) $\{x \mid x \neq 2, x \neq -7, x \neq -3\}$

C) $\{x \mid x \neq 2, x \neq -7, x \neq 3\}$

D) $\{x \mid -7 < x < -3 \text{ and } x > 2\}$

E) $\{x \mid -7 < x < -3 \text{ and } x \leq 2\}$

7) What is the Domain of the following function?

$$g(x) = \frac{10}{8-x}$$

A) $\{x \mid x \in \mathbb{R}\}$

B) $\{x \mid x < 10\}$

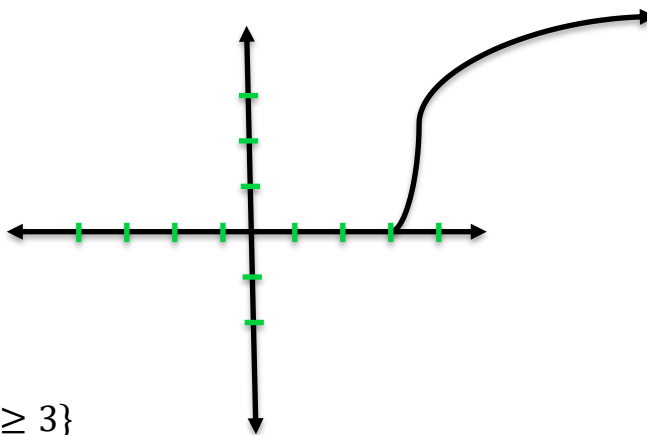
C) $\{x \mid x < 8\}$

D) $\{x \mid x \neq 8\}$

E) $\{x \mid x = 2\}$

1	2	3	4	5	6	7	8	9	10
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8) What is the Domain and Range of the following function?

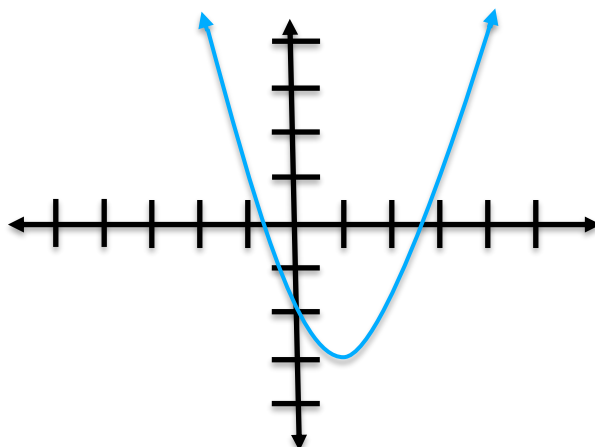


- A) Domain; $\{x|x \geq 3\}$
Range; $\{y|y \geq 0\}$
- B) Domain; $\{x|x \in \mathbb{R}\}$
Range; $\{y|y \geq -3\}$
- C) Domain; $\{x|x \in \mathbb{R}\}$
Range; $\{y|y \geq 3\}$
- D) Domain; $\{x| -3 < x < 3\}$
Range; $\{y|y > 3\}$
- E) Domain; $\{x| -3 < x < 3\}$
Range; $\{y|y \in \mathbb{R}\}$

9) Which of the following is a set of points on a line with an **undefined slope**?

- A) $\{(3, 7), (4, 8)\}$
- B) $\{(-1, 7), (3, 7)\}$
- C) $\{(-2, 4), (-3, 4)\}$
- D) $\{(1, 3), (1, -10)\}$
- E) $\{(3, 5), (5, 3)\}$

10)



What is the **Range** of the graph above?

- A) $\{y|y \geq -3\}$
- B) $\{y|y \geq -2\}$
- C) $\{y|y < 3\}$
- D) $\{y|y > -2\}$
- E) $\{y|y \geq 2\}$

11) Which value/values of x make the following function **undefined**?

$$\frac{1}{(x-3)(x+4)(x-5)}$$

- A) 3, -4, 5
- B) -3, 4, -5
- C) 0, 3
- D) 5
- E) -60

12) Every line has a range of **all real numbers except**?

- A) A horizontal Line
- B) A vertical Line
- C) A line parallel to the line $y = x$
- D) A line parallel to the line $y = -x$
- E) A line perpendicular to the x -axis

Solutions Chapter 31

Domain Range Undefined

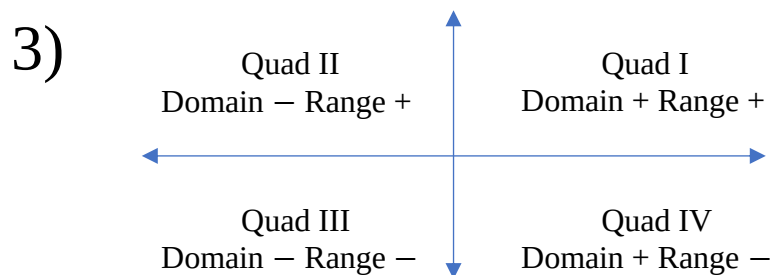
- | | |
|------|-------|
| 1) E | 7) D |
| 2) D | 8) A |
| 3) C | 9) D |
| 4) C | 10) A |
| 5) C | 11) A |
| 6) D | 12) A |

1) Two things make a function undefined; a negative value under a radical with an even root (like a square root) or a denominator with zero value. In this question, the radical must be zero or positive since it is the square root or second root which is even. Remember it is ok to take the square root of zero; it is equal to zero. Therefore, $x - 7 < 0$; $x < 7$ are values that make this function undefined. Any value less than 7 is undefined. The only possible answer choice that is not less than 7 and is not undefined is 7.

7 Answer E Difficulty Level 5

2) Two things make a function undefined; a negative value under a radical with an even root (like a square root) or a denominator with zero value. In this question, the denominator will equal zero if $x - 5 = 0$ or $x = 5$, which would make the function undefined.

5 Answer D Difficulty Level 2.5



Only quadrant II (2) has a Range with all positive numbers and a Domain with all negative numbers.

II Only Answer C Difficulty Level 6.5

4) Two things make a function undefined; a negative value under a radical with an even root (like a square root) or a denominator with zero value. In this question, the denominator will equal zero if $x = 2$. Therefore, 2 is not in the domain of x . Looking more closely, the sign under the radical must be greater than or equal to zero. The numerator is a trinomial in the form $ax^2 + bx + c$. Always look to factor these on the Math ACT. $x^2 - 5x + 6$ can be factored (two numbers that multiply to 6 and add to -5); -3 and -2; $(x - 3)(x - 2)$.

The equation simplifies to $\sqrt{\frac{(x-3)(x-2)}{(x-2)}} = \sqrt{x-3}$ Since there cannot be a negative number under a square root sign, $x - 3 \geq 0$, $x \geq 3$.

Therefore, $x \neq 2$ and $x \geq 3$. Since 2 is less than 3, the domain is $x \geq 3$.

$x \geq 3$ Answer C Difficulty Level 7

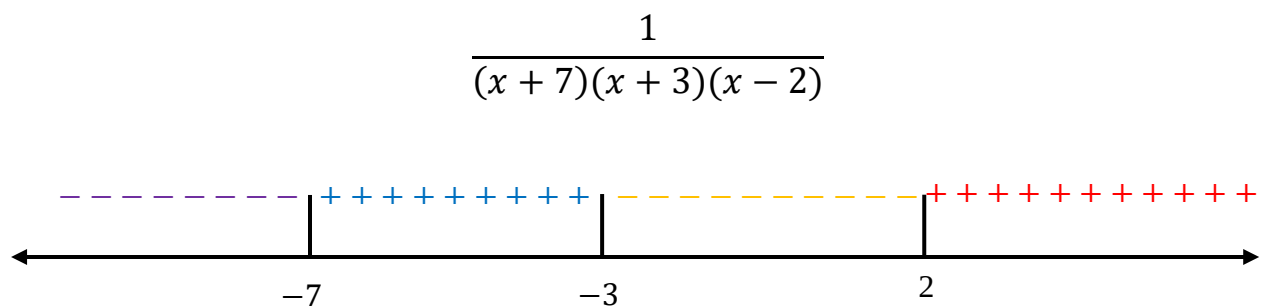
5) Two things make a function undefined; a negative value under a radical with an even root (like a square root) or a denominator with zero value. Checking each answer choice; A) $3x - 2$ is an equation of a line with a slope of 3 and a domain of all real numbers, B) $\frac{12}{x+7}$ has a domain of all real numbers except -7, because that would make the denominator of the fraction zero, (note; $x = -7$ is the vertical asymptote of the function and is not part of the domain),

C) $\frac{12}{x-7}$ has a domain of all real numbers except 7, because that would make the denominator of the fraction zero, (note; $x = 7$ is the vertical asymptote of the function and is not part of the domain), D) $\frac{1}{x+7}$ has a domain of all real numbers except -7, because that would make the denominator of the fraction zero, (note; $x = -7$ is the vertical asymptote of the function and is not part of the domain) and E) $3x + 2$ is an equation of a line with a slope of 3 and a domain of all real numbers.

$\frac{12}{x-7}$ Answer C Difficulty Level 3

6) Two things make a function undefined; a negative value under a radical with an even root (like a square root) or a denominator with zero value. Any number that makes the denominator equal to zero is not in the function's domain, therefore $x + 7 \neq 0$ or $x \neq -7$, $x + 3 \neq 0$ or $x \neq -3$, and $x - 2 \neq 0$ or $x \neq 2$. The numbers -7, -3, and 2 are excluded from the domain because they make the denominator zero. Next, check which numbers or intervals make the full expression under the radical negative as negative numbers under the radical must also be excluded from

the domain. Finally, use the three integers that made the denominator zero as checking points to see where the function under the radical is negative;



Start by choosing **any number less than -7** to see if the left end of the number line is negative (not in the domain) or positive (in the domain).

Choosing -10 for x ; $\frac{1}{((-10)+7)((-10)+3)((-10)-2)} = \frac{1}{(-3)(-7)(-12)}$, the values inside the parenthesis do not matter; only the signs are important $\frac{1}{(-)(-)(-)}$ makes the expression negative, since a negative number times a negative number times a negative number is **negative** then $\frac{1}{-}$ is negative (**shown on the number line above**); Next, check the interval from -7 to -3 by choosing any number in between them. Choosing -5 , $\frac{1}{(2)(-2)(-7)}$ or $\frac{1}{(+)(-)(-)}$ makes the expression positive because a positive times a negative times a negative equals a **positive** (**shown on the number line above**); Continue the same process for the last two intervals; Check the interval from -3 to 2 by choosing any number in between them. Choosing 0 , $\frac{1}{(+)(+)(-)}$ makes the expression negative because a positive times a positive times a negative equals a **negative**, (**shown on the number line above**). Note: always use zero if it is in the interval being tested as it is the fastest. Finally, check the right-hand side of the number line to see if numbers greater than 2 are in the domain; Use any number greater than two; choose 3 for x to get the following signs $\frac{1}{(+)(+)(+)}$ which is a **positive** number **shown in the number line above**. Finally, the domain is every value in the number line that is positive, excluding the three checking points.

$$\{x \mid -7 < x < -3 \text{ and } x > 2\}$$

Answer D Difficulty Level 8

7) Two things make a function undefined; a negative value under a radical with an even root (like a square root) or a denominator with zero value. The only value of x that would make this function undefined makes the denominator equal to zero. Set

the denominator equal to zero; $8 - x = 0$ and solve, $x = 8$. Therefore, 8 is the only value excluded from the domain.

$\{x|x \neq 8\}$ Answer D Difficulty Level 2

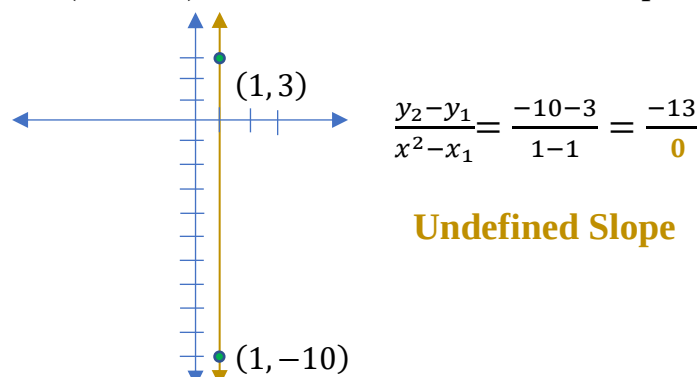
8) The function's domain is the set of all x-values of the function, and the range is the set of all y-values of the function. When finding the domain, always starts from the far left(negative infinity) on the graph then move to the right. Starting from negative infinity and moving to the right, the first x-value on the graph is when $x = 3$. The graph continues in the positive direction without bound towards positive infinity; therefore, the domain is from 3 to infinity and includes 3. The domain is x is greater than or equal to 3; $x \geq 3$. When finding the range of a function, start from the bottom(negative infinity) of the graph then move upward. The first y-value on this graph is on the x-axis where $y = 0$. As the y-values increase, the graph will continue to increase without bound.

The range is from $y = 0$ to infinity; $y \geq 0$

Domain: $\{x|x \geq 3\}$ Range: $\{y|y \geq 0\}$

Answer A Difficulty Level 5

9) In math, a vertical line is a line with an undefined slope. One way to check if a line has an undefined slope is by graphing the coordinates. Another way to determine which set of points connects a line with an undefined slope is to pick any two points on the line and find the slope using the slope formula, $\frac{y_2 - y_1}{x_2 - x_1}$. If the formula produces a zero in the denominator, the line has an undefined slope. Finally, the easiest way to find which coordinates determine the line with an undefined slope is to find the line containing **two of the same x-values** but **two different y-values**. Since **(1, 3)** and **(1, -10)** fit the criteria, it is the set of points on a line with an undefined slope.



$\{(1, 3), (1, -10)\}$ Answer D Difficulty Level 2.5

10) The range of a function is the function's y-values. When finding the range, while analyzing a graph, start from the bottom (negative infinity) and move upward. In this graph, starting from negative infinity and moving upward, the first y-value on the graph is -3, and the graph continues upward without bound. The graph includes -3.

$\{y|y \geq -3\}$ **Answer A** Difficulty Level 4

11) Any value of x that makes the denominator equal to zero is undefined. If $x = 3, -4,$ or $5,$ the denominator will equal zero because zero times anything equals zero (multiplication property of zero);

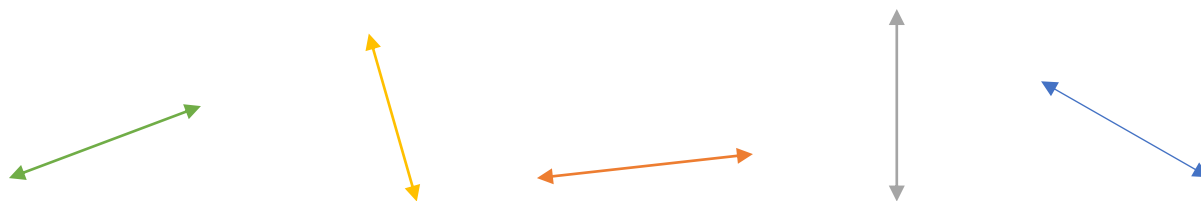
if $x = 3; \frac{1}{(3-3)(3+4)(3-5)} = \frac{1}{(0)(7)(-2)} = \frac{1}{(0)}$; undefined

if $x = -4; \frac{1}{(4-3)(-4+4)(-4-5)} = \frac{1}{(-7)(0)(-9)} = \frac{1}{(0)}$; undefined

if $x = 5; \frac{1}{(5-3)(5+4)(5-5)} = \frac{1}{(2)(9)(0)} = \frac{1}{(0)}$; undefined

3, -4, and 5 **Answer A** Difficulty Level 3.5

12) Since lines go on forever, every line will have a range of all real numbers as each will eventually reach negative infinity as the y-values decrease and eventually reach positive infinity as the y-values increase except one line;



The horizontal line is the only line that the y-values never increase or decrease and thus do not have a range of all real numbers. A horizontal line has a range equal to only one real number.



A Horizontal Line **Answer A** Difficulty Level 6